

National College of Ireland
Cloud Competency Centre

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Lab: Hadoop 101

Cloud Architectures (CLAR) : MSc Cloud Computing : Prof-Dr. H. González–Vélez
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Contents

Overview **2**

Prerequisites **2**
Install Java . . . 2
Create a new user and group to run Hadoop . . . 2
Set up passphrase-less SSH . . . 2
Disable IPv6 . . . 3

Install Hadoop **3**
Download Hadoop . . . 3
Verify the downloaded archive file . . . 4
Unpack the downloaded archive file . . . 4
Set permissions and create a symbolic link . . . 4

Configure Hadoop **4**
Set the environmental variable for the Java home directory . . . 6
Create a data directory for HDFS . . . 6
Format the HDFS namenode . . . 7
Starting Hadoop services . . . 7

Testing Hadoop - running a word count job **7**
Did we produce anything? . . . 8

Overview

In this lab we will install and configure a **Hadoop** instance and perform a sample **Map-Reduce** task on a text from Oscar Wilde. We will configure Hadoop in **pseudo-distributed** mode. In other words, we will run Hadoop on a single computer but with each of the Hadoop instance running in a separate Java process, simulating a true distributed configuration.

Prerequisites

This lab presumes that you have already instantiated an Ubuntu virtual machine in AWS (or OpenStack). Start the virtual machine and log in using the username you created when you first built the virtual machine.

Install Java

At the terminal (console) of the virtual machine, execute each of the following commands in turn. Press *Enter* after typing each command. This will apply to all commands entered at the terminal.

```
sudo apt update
sudo apt -y upgrade
sudo apt install openjdk-8-jdk
```

If prompted, type *y* to confirm that you wish to install the packages. Note that the command **sudo** runs any commands following it with superuser privileges. **N.B.** Newer versions of Java should work but you will need to test accordingly.

Create a new user and group to run Hadoop

Enter the command below to add a new user-group to your system. The group is called *hadoop*.

```
sudo groupadd hadoop
```

Next create a new user, adding it to the group we just created. The username is *hduser*.

```
sudo useradd -ghadoop hduser -m -s /bin/bash
```

The newly-created user will need to have a password set. The command below will set the password for the user hduser. When prompted, enter the password *hadoop* and confirm it. **Do not change this password**, as it will help ensure that your use of the VM can be easily supported during lab sessions.

```
sudo passwd hduser
```

Finally, the newly-created user will need to be able to avail of superuser privileges. Run the following command to add the user to the group with those privileges.

```
sudo usermod -aG sudo hduser
```

Set up passphrase-less SSH

You will be setting up Hadoop to run in pseudo-distributed mode. This is where Hadoop launches separate process to imitate running across multiple interconnected nodes. In a distributed environment, In a distributed environment Hadoop uses the *secure shell* protocol (SSH) to communicate with and manage its nodes.
We first need to log in to the **hduser** account. Do this by typing the following command.

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```
su - hduser
```

The next step is to create a SSH (passphrase-less) key pair for the hduser account.

```
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa
```

The newly generated keys are then copied to hduser's authorised keys using the command below.

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

In order to ensure that everything went well, we test that passphrase-less SSH works using the following command. The first time you run this command you will be asked to accept the key fingerprint. Do so and you should find yourself back at the prompt.

```
ssh localhost
```

At this point you should be logged in to the hduser account using SSH. You should now exit from the SSH shell session using the command below.

```
exit
```

Note: If are prompted for a password when executing the `ssh localhost` command then SSH is not configured correctly. You should revisit all steps in this section until SSH works as expected.

Disable IPv6

Hadoop does not currently run over IPv6 networks, so we need to disable IPv6 using the command below.

```
sudo nano /etc/sysctl.conf
```

This will open a text editor. If you are a Linux pro you can, of course, use your favourite editor instead (e.g. vi). Add the following entries at the end of the file and then save the file by pressing Ctrl and O simultaneously.

```
net.ipv6.conf.all.disable_ipv6=1
net.ipv6.conf.default.disable_ipv6=1
net.ipv6.conf.lo.disable_ipv6=1
```

Exit the editor by pressing *Ctrl* and *X* simultaneously.
To make these changes take immediate effect, run the following command:

```
sudo sysctl -p
```

Install Hadoop

Download Hadoop

We will use the current version of the 3.4 branch, which at the time of writing (2nd December 2024) is 3.4.1. To download the Hadoop archive, type the following commands in the terminal and press *Enter* after each one.

```
cd ~
curl -O http://ftp.heanet.ie/mirrors/www.apache.org/dist/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz
```

If *curl* is not found on your system, you may need to install it with the command below:

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```
sudo apt install curl
```

Verify the downloaded archive file

Checking the integrity of the downloaded archive is optional but **highly recommended**.* Do do this, enter the following commands.

```
cd ~
curl -O https://www.apache.org/dist/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz.sha512
shasum -a 512 ./hadoop-3.4.1.tar.gz
```

Compare the output of this command with the line beginning with *SHA512=* in the file `hadoop-3.4.1.tar.gz.sha512`. If the two match, the integrity of the archive can be assured. Note that you can open the *mds* file with any text editor.

Unpack the downloaded archive file

We will now change the working directory to where we want to install Hadoop by entering the following into the terminal and pressing Enter. This is where we will install all downloaded software.

```
cd /usr/local
```

Next we unpack the archive we downloaded earlier. Type the two lines below into the terminal, pressing Enter after each as usual. The second line removes (deletes) the downloaded archive file as it is no longer needed.

```
sudo tar xvf ~/hadoop-3.4.1.tar.gz
rm ~/hadoop-3.4.1.tar.gz
```

The command tar followed by *x* unpacks an archive, much in the same way as *zip* archives are extracted on Windows or MacOS. The tilde ~ is an alias for the current user's home directory.

Set permissions and create a symbolic link

Create a symbolic link (a form of shortcut) to alias the directory `/usr/local/hadoop` with `/usr/local/hadoop-3.4.1` This allows us to potentially have multiple versions of Hadoop installed and to switch between the by simply deleting and recreating the symbolic link.

```
sudo ln -s hadoop-3.4.1 hadoop
```

Last of all, we run a command to ensure that all files in the hadoop directories are owned by the user *hduser* and the group *hadoop*.

```
sudo chown -R hduser:hadoop hadoop*
```

The command **chown** changes the ownership of a file or directory. The asterisk after the word hadoop means that it will operate on any file or directory starting with that word.

Configure Hadoop

First change to the directory where the Hadoop configuration files are stored. This can be done either by issuing a change directory command relative to the current directory (`/usr/local`) as shown below:

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```
cd hadoop/etc/hadoop
```

or by entering the full path (an absolute path) as in the following example.

```
cd /usr/local/hadoop/etc/hadoop
```

Either one will do the job, but you will find relative paths easier to type, provided you know where the destination directory is relative to your current directory.
Having changed directory, the first task is to ensure Hadoop knows where to find the Java Virtual Machine.
First make a copy of the MapReduce configuration template.

```
cp mapred-site.xml mapred-site.xml.template
```

Next a text editor to edit the `mapred-site.xml` file. Here we are using *nano* but you can use any other suitable text editor.

```
nano mapred-site.xml
```

Edit the file to ensure the following configuration entries are present and then save the file.

```
<configuration>
    <property>
        <name>mapreduce.jobtracker.address</name>
        <value>local</value>
    </property>
</configuration>
```

Edit the `core-site.xml` file with the following configuration entries and save the file.

```
<configuration>
    <property>
        <name>hadoop.tmp.dir</name>
        <value>/home/hduser/tmp</value>
    </property>
    <property>
        <name>fs.defaultFS</name>
        <value>hdfs://localhost:54310</value>
    </property>
</configuration>
```

Edit the `hdfs-site.xml` file with the following configuration entries and save the file.

```
<configuration>
    <property>
        <name>dfs.replication</name>
        <value>1</value>
    </property>
    <property>
        <name>dfs.datanode.data.dir</name>
        <value>/home/hduser/hdfs</value>
    </property>
</configuration>
```

Set the environmental variable for the Java home directory

First of all, you will need to identify the directory where Java is installed, by running the command below:

```
sudo update-alternatives --config java
```

You should have only one entry for java. Take a note of this, taking care to remove the trailing `/bin/java` or `jre/bin/java`. So the following path returned by the command.

`/usr/lib/jvm/java-8-openjdk-amd64/bin/java`

should be recorded as:

`/usr/lib/jvm/java-8-openjdk-amd64`

Next use a text editor to edit the `hadoop-env.sh` file, changing the line starting with export `JAVA_HOME=`, putting the path to your Java installation immediately after the equals symbol and then save the file. For the example above, this would be:

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
```

Note that the version of Java installed on your virtual machine may be different!

Create a data directory for HDFS

In the previous section, we specified the locations of two directories where Hadoop will store its files. We now need to create them and set the access permissions for one of the two.

```
mkdir ~/tmp
mkdir ~/hdfs
chmod 750 ~/hdfs
```

The command **chmod** changes the access permissions for a file or directory. ((In octal mode, it takes the following arguments:

```
chmod <options> <octalmode> <fileordir>
```

where *octalmode* is a three digit number, the leftmost digit indicates the permissions for the user, the middle digit for the group and the rightmost for all other users. Each digit can take a value between 0 and 7. In octal mode:

- 4 stands for **read**;
- 2 stands for **write**,
- 1 stands for **execute**; and
- 0 stands for **none**.

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This can be combined to give a range of values from 0 to 7. The meaning of each value is given in the table on the next page.

Value	Permissions	Symbolic
7	read, write and execute	rwx
6	read and write	rw-
5	read and execute	r-x
4	read only	r–
3	write and execute	-wx
2	write only	-w-
1	execute only	–x
0	none	—

Note that the symbolic values in the table are a reference to another method of supplying parameters to the *chmod* command. This is left up to you to investigate!

Note: Be very careful when using either *chmod* or *chown*. You should always confirm which directory you are in before issuing either command as it is entirely possible to render your installation unusable if these commands are issued in the wrong directory (for instance, in `/usr/local` rather than in `/usr/local/hadoop`).

Format the HDFS namenode

We will now format the HDFS namenode.

```
cd /usr/local/hadoop
bin/hdfs namenode -format
```

Starting Hadoop services

```
sbin/start-dfs.sh
sbin/start-yarn.sh
```

Using the Java Process Status tool (JPS) we can check that everything is running.

```
jps
```

Testing Hadoop - running a word count job

In order to carry out a word count, some text is needed. Download the text of Oscar Wilde's "The Importance of being Earnest" from the Project Gutenberg site.

```
cd /tmp
mkdir gutenberg
cd gutenberg
wget https://www.gutenberg.org/files/844/844-0.txt
```

We first copy the text file from the local filesystem to HDFS.

```
cd /usr/local/hadoop
bin/hdfs dfs -mkdir /user # make a directory in HDFS
bin/hdfs dfs -mkdir /user/hduser
```

```
bin/hdfs dfs -mkdir /user/hduser/gutenberg
bin/hdfs dfs -copyFromLocal /tmp/gutenberg/* /user/hduser/gutenberg
```

Then run an example Map-Reduce task (wordcount). **Note:** The commands below are spread across 2 lines to facilitate printing. They should be combined into one single command.

```
bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-3.4.1.jar \
wordcount /user/hduser/gutenberg /user/hduser/gutenberg-output
```

Did we produce anything?

We can first check the output directory on HDFS.

```
bin/hdfs dfs -ls /user/hduser/gutenberg-output
```

If the wordcount operation worked, there should be a file named `part-r-00000`. This should be downloaded from HDFS to the local file system as follows:

```
mkdir /tmp/gutenberg-output
bin/hdfs dfs -get /user/hduser/gutenberg-output/part-r-00000 /tmp/gutenberg
```

Inspect the output file using nano or another text editor.

```
nano /tmp/gutenberg-output/part-r-00000
```

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